# CV

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**Data of Birth:** 1/1/1984
**Work:** School of Computer Science and Information Engineering, Hefei University of Technology (HFUT)
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## Research Interests

Network measurement, Deep learning, Anomaly detection, Pattern recognition

## Education Background

2007.09-2012.07, Ph.D., State Key Laboratory of Networking and Switching Technology, Beijing University of Posts and Telecommunications, China, supervised by Prof. Luoming Meng
2003.09-2007.06, Bachelor, Information and Computing Science, Hebei Normal University, China

## Work Experiences

2021.01-Now, Hefei University of Technology, School of Computer Science and Information Engineering, **Associate Professor**, **PhD Supervisor**
2019.08-2020.08, University Of Victoria, Canada, **Visiting Scholar**, worked with Prof. Kui Wu
2017.09-2017.12, Dresden University of Technology (in Germany), Faculty of Education, **Visiting Scholar**
2012.08-2013.08, School of Computer Engineering, Nanyang Technological University, **Postdoc Research Fellow**, worked with Jun Luo (IEEE Fellow)

## Scientific Research Projects

2026.01–2029.12, **National Natural Science Foundation of China (NSFC) – General Program**, No. 62572168, "Research on Self-evolving Attack Detection Mechanisms in Dynamic Network Environments", 500,000 CNY.
2015.01–2017.12, **NSFC – Youth Program**, No. 61402013, "Research on Link Loss Inference Mechanism based on Active Probing for Uncertain IP Networks", 260,000 CNY.
2025.09–2028.08, **Anhui Provincial Natural Science Foundation – General Program**, No. 2508085MF151, "Research on Covert Abnormal Traffic Detection Methods in Uncertain Networks", 80,000 CNY.
2020.07–2023.06, **Anhui Provincial Natural Science Foundation – General Program**, No. 2008085MF203, "Research on Anomaly Detection Mechanism of Agricultural IoT Data Based on Deep Learning", 120,000 CNY.
2018.07–2021.06, **Open Project of State Key Laboratory of Networking and Switching Technology**, No. SKLNST-2018-1-10, "Research on Anomaly Detection Mechanisms for Large-scale Irregular Sensor Networks".
2017.01–2018.12, **Open Project of State Key Laboratory of Networking and Switching Technology**, No. SKLNST-2016-1-02, "Research on Link Loss Inference Mechanisms for Large-scale Uncertain Networks".

## Academic Memberships and Positions

**Memberships**
IEEE Member
IEEE Signal Processing Society (SPS) Member
IEEE Computer Society (CS) Member
IEEE Communications Society (ComSoc) Member
CCF Senior Member
Executive Member, CCF Internet Committee
Executive Member, CCF Network and Data Communication Committee

**Associate Editor**
2024-2025, IEEE Transactions on Network and Service Management (TNSM)
2024-Now, Computer Networks

**Conference Program Committee/ Session Chair**
The 20th ACM International Conference on Web Search and Data Mining (WSDM 2027) (Program Committee)
The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026) (Ses-

sion Chair)
The 37th IEEE International Conference on Computer Communications (INFOCOM 2018) (Session Chair)

**Reviewer for Transactions, Journals, and Magazines**
IEEE Transactions on Information Forensics and Security (TIFS)
IEEE/ACM Transactions on Networking (ToN)
IEEE Transactions on Knowledge and Data Engineering (TKDE)
IEEE Internet of Things Journal
IEEE Signal Processing Letters
ACM Transactions on Privacy and Security (TOPS)
IEEE Transactions on Services Computing (TSC)
Journal of Systems Architecture (JSA)
Wireless Communications and Mobile Computing

**Reviewer for Conferences**
Annual Conference on Neural Information Processing Systems (NeurIPS 2025) (Top Reviewer)
International Conference on Learning Representation (ICLR 2025)
International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025)
IEEE International Conference on Computer Communications (INFOCOM 2023)
IEEE Symposium on Computers and Communications (ISCC 2022)
IEEE/ACM International Symposium on Quality of Service(IWQoS 2021)
International Wireless Communications and Mobile Computing (IWCMC 2021)

## Publications

**Journals and Magazines**

(1) Y. Qiao, J. Wang, M.Li, R, Hu, Z. Hu, C. Feng, W. Li, M. Li, M. Conti, "ImConDM: Accurate MTS Anomaly Detection Integrating Imputation with Conditional Diffusion Models". ***IEEE Transactions on Information Forensics and Security (TIFS)***, 2026 (Online).

(2) M. Li, Y. Chen,Y. Qiao*, G. Ye, Z. Zhang, L. Zhu, M. Conti, "Trust in a Decentralized World: Data Governance From Faithful, Private, Verifiable, and Traceable Data Feeds," ***IEEE Transactions on Information Forensics and Security (TIFS)***, 2025, 20: 10818–10833.

(3) M. Li, H. Ding, Y. Chen, Y. Qiao*, Z. Zhang, L. Zhu, M. Conti, "Threshold Signatures With Verifiably Timed Combining and Message-Dependent Tracing," ***IEEE Transactions on Information Forensics and Security (TIFS)***, 2025, 20: 9477–9491.

(4) Y. Qiao, M. Li, X. Yuan, K. Wu, C. Feng, M. Li, K. Xie, "Routing-Oblivious and Data-Efficient Network Tomography With Flow-Based Generative Model," ***IEEE/ACM Trans. Networking (TON)***, 2026, 34: 3505-3520.

(5) X. Yuan, Y. Qiao, M. Li, Z. Wei, C. Feng, Z. Wang, W. Chen, "Learning-based Sketches for Frequency Estimation in Data Streams without Ground Truth," ***IEEE Trans. Knowledge and Data Engineering (TKDE)***, 2026, 38 (6): 3722–3736.

(6) Y. Qiao, R. Hu, P. Zhao, J. Wang, Z. Wei, X. Ma, W. Li, "3DDPS: A Traffic Matrix Estimation Method based on Three-dimensional Diffusion Posterior Sampling," ***Computer Networks***, 2025, 257 (6): 111007.

(7) Y. Qiao, K. Wu, X. Yuan, "AutoTomo: Learning-Based Traffic Estimator Incorporating Network Tomography," ***IEEE/ACM Transactions on Networking (ToN)***, 2024, 32 (6): 4644-4659.

(8) Y. Qiao, K. Wu, P. Jin, "Efficient Anomaly Detection for High-Dimensional Sensing Data with One-Class Support Vector Machine," ***IEEE Transactions on Knowledge and Data Engineering (TKDE)***, 2023, 35 (1): 404-417.

(9) Y. Qiao, K. Wu, M. Khabbazian, "Non-Intrusive Balance Tomography Using Reinforcement Learning in the Lightning Network," ***ACM Transactions on Privacy and Security (TOPS)***, 2024, 1 (27): 1-32.

(10) Y. Qiao, J. Jiao, X. Cui, Y. Rao, "Robust Loss Inference in the Presence of Noisy Measurements & Hidden Fault Diagnosis," ***IEEE/ACM Transactions on Networking (ToN)***, 2020, 28 (1): 43-56.

(11) X. Yuan, Y. Qiao*, Z. Wei, Z. Zhang, M. Li, P. Zhao, R. Hu, W. Li, "Diffusion Models Meet Network Management: Improving Traffic Matrix Analysis with Diffusion-based Approach," ***IEEE Trans. Network and Service Management (TNSM)***, 2025, 22 (2): 1259–1275.

(12) Y. Qiao, P. Zhao, J. Wang, et al., "Multivariate Time Series forecasting based on temporal decomposition and graph neural network," ***Engineering Applications of Artificial Intelligence***, 2025, 161: 112074.

(13) K. Wang, Y. Qiao, N. Xing, "How to Couple Two Networks for a Smart Grid", ***IEEE Access***, 2018, 6: 34643 - 34650.

(14) Y. Qiao, J. Jiao, H. Ma, "Adaptive path selection for link loss inference in network tomography applications," ***Plos One***, Oct. 2016.

(15) Z. Hu, Y. Qiao, J. Luo, "ATME: Accurate Traffic Matrix Estimation in both Public and Private Datacenter Networks," ***IEEE Transactions on Cloud Computing (TCC)***, 2018, 6 (1): 60-73.

(16) Z. Hu, Y. Qiao, J. Luo, "Coarse-Grained Traffic Matrix Estimation for Data Center Networks," ***Computer Communications***, 2015, 56: 25-34.

(17) Y. Qiao, X. Cui, P. jin, "Fast outlier detection for high-dimensional data of wireless sensor networks," ***International Journal of Distributed Sensor Networks (IJDSN)***, 2020, 16(10).

(18) Y. Qiao, L. Cheng, X. Qiu, L. Meng, R. Gu, "Efficient loss inference algorithm with unicast end-to end measurements," ***Journal of Network and Systems Management (JNSM)***, 2013, 21: 169-193.

(19) Y. Qiao, X. Qiu, L. Meng, "Efficient Probe Selection for Fault Localization Using the Property of Submodularity," ***International Journal of Communication Systems (IJCS)***, 2013, 26(1): 84-99.

(20) Y. Qiao, X. Qiu, L. Meng, "Accurate diagnosis in computer networks using unicast end-to-end measurements," ***IEICE TRANSACTIONS on Communications (IEICE T COMMUN)***, 2013, E96(2): 522-532.

**Conferences**

(1) X. Yuan, Y. Qiao*, Z. Wang, W. Chen, "LMTE: Putting the "Reasoning" into WAN Traffic Engineering with Language Models," ***Proc. IEEE INFOCOM***, May 2026, Tokyo Japan.

(2) F. Yan, K. Jiang, Y. Qiao*, M. Li, Y. Li, P. Yu, C. Feng, "ARI-LLM: Autoregressive Imputation for Network Traffic Matrix via Large Language Models," ***Proc. IEEE INFOCOM***, May 2026, Tokyo Japan.

(3) X. Yuan, Y. Qiao, Z. Wang, W. Chen*, "Divide, Harmonize, Then Conquer It: Shooting Multi-Commodity Flow Problems with Multi-Modal Language Models," ***Proc. ICLR***, 2026, Rio de Janeiro, Brazil.

(4) Y. Qiao, J. Song, J. Dong, et al., "C2FFormer: Coarse-to-Fine Time Series Imputation via Autoregressive Transformer," ***Proc. PAKDD***, Jun. 2026, Hong Kong, China.

(5) Y. Qiao, X. Yuan, K. Wu "Routing-Oblivious Network Tomography with Flow-Based Generative Model," ***Proc. IEEE INFOCOM***, May 2024, Vancouver, Canada.

(6) Y. Qiao, "Robust Loss Inference in the Presence of Noisy Measurements," ***Proc. IEEE INFOCOM***, Apr. 2018, Honolulu, HI, USA.

(7) X. Yuan, Y. Qiao*, "Diffusion-TS: Interpretable Diffusion for General Time Series Generation, International Conference on Learning Representations", ***Proc. ICLR***, May 2024, Vienna Austria.

(8) Z. Hu, Y. Qiao*, Z. Cai, et al., "Dual Trajectory Revised Diffusion Model for Time Series Forecasting," ***Proc. IEEE ICASSP***, Apr. 2025, Hyderabad, India.

(9) K. Jiang, F. Yan, Y. Qiao, M. Li, Y. Li, M. Conti, "Network Traffic Matrix Imputation via Large Language Models," ***Proc. IEEE ISCC***, Jul. 2025, Bologna, Italy.

(10) R. Hu, X. Yuan, Y. Qiao*, B. Zhang, P. Zhao, "Unsupervised Anomaly Detection For Multivariate Time Series Using Diffusion Model," ***Proc. IEEE ICASSP***, Apr. 2024, Seoul, Korea.

(11) Y. Qiao, K. Wu, K. Majid, "Non-Intrusive and High-Efficient Balance Tomography in the Lightning Network," ***Proc. ACM ASIACCS***, Jun 2021, Hong Kong, China.

(12) X. Yuan, Y. Qiao*, P. Zhao, R. Hu, B. Zhang, "Traffic Matrix Estimation based on Denoising Diffusion Probabilistic Model," ***Proc. IEEE ISCC***, Jul. 2023, Gammarth, Tunisia.

(13) Y. Qiao, Z. Hu, J. Luo, "Efficient Traffic Matrix Estimation for Data Center Networks," ***Proc. IFIP Networking*** (Best paper nominated), May 2013, New York, USA.

(14) X. Yu, Y. Ye， J. Wang, Y. Qiao*,"Practical Loss Inference in Uncertain Networks," ***Proc. IEEE ISCC***, Jul 2017, Heraklion, Crete, Greece.

(15) Z. Hu, Y. Qiao, J. Luo,"CREATE: CoRrelation enhanced trAffic maTrix estimation in data center networks," ***Proc. IFIP Networking***, Jun. 2014, Trondheim, Norway.

(16) Y. Qiao, G. Wang, X. Qiu, R. Gu, "Network Loss Tomography Using Link Independence," ***Proc. IEEE ISCC***, Jul. 2012, Cappadocia, Turkey.

(17) Y. Qiao, L. Cheng, X. Qiu, L. Meng, 'A Methodology Used to Optimize Probe Selection," ***Proc. IEEE Globecom***, Dec. 2010, Miami, FL, USA.

(18) L. Cheng, X. Qiu, L. Meng, Y. Qiao, R. Boutaba, 'Efficient Active Probing for Fault Diagnosis in Large Scale and Noisy Networks," ***Proc. IEEE INFOCOM***, May 2010, San Diego, CA, USA.

## Awards and Honors

2013.07, Excellence in Doctoral Thesis Award of Beijing University of Posts and Telecommunications

2018.05, Best Presentation in Section of INFOCOM 2018